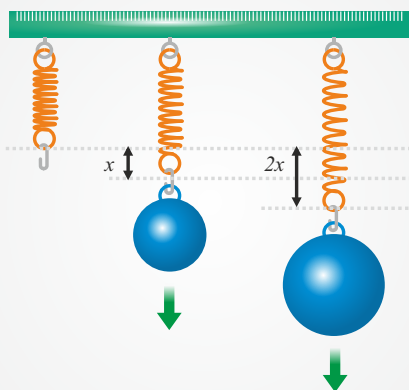


PHYSICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Study Material

**Elasticity, Thermal Expansion,
Calorimetry & Heat Transfer**

(English Medium)

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ELASTICITY, THERMAL EXPANSION, CALORIMETRY & HEAT TRANSFER**ELASTICITY****STRESS**

The internal restoring force acting per unit area of cross-section of the deformed body is called stress.

$$\text{Stress} = \frac{\text{Internal restoring force}}{\text{Area of cross section}} = \frac{F_{\text{internal}}}{A} = \frac{F_{\text{external}}}{A}$$

Stress depends on direction of force as well as direction of area of application so it is tensor.

SI UNIT : N-m^{-2}

Dimensions : $\text{M}^1 \text{L}^{-1} \text{T}^{-2}$

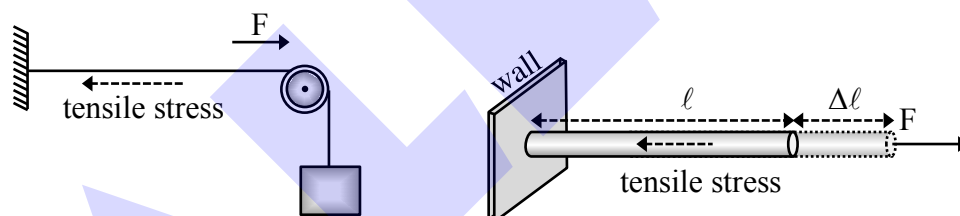
There are three types of stress :-

Longitudinal Stress :

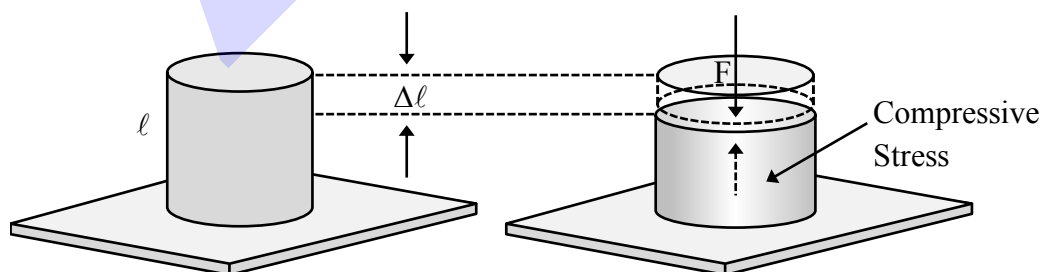
When the stress is normal to the surface of body, then it is known as longitudinal stress. There are two types of longitudinal stress

(a) Tensile Stress

The longitudinal stress, produced due to increase in length of a body, is defined as tensile stress.

**(b) Compressive Stress**

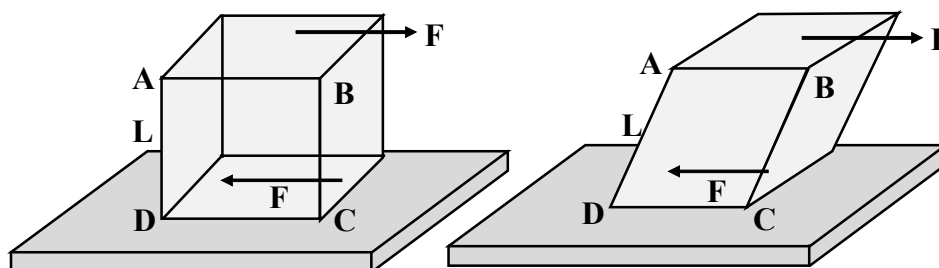
The longitudinal stress, produced due to decrease in length of a body, is defined as compressive stress.

**(c) Volume Stress**

If equal normal forces are applied every one surface of a body, then it undergoes change in volume. The force opposing this change in volume per unit area is defined as volume stress.

(d) Tangential Stress or Shear Stress

When the stress is tangential or parallel to the surface of a body then it is known as shear stress. Due to this stress, the shape of the body changes or it gets twisted.

**STRAIN**

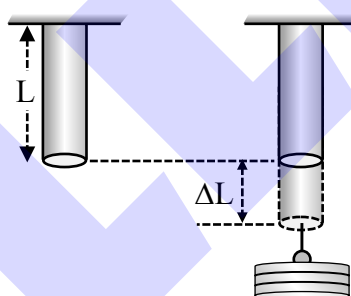
The ratio of change of any dimension to its original dimension is called strain.

$$\text{Strain} = \frac{\text{change in size of the body}}{\text{original size of the body}}$$

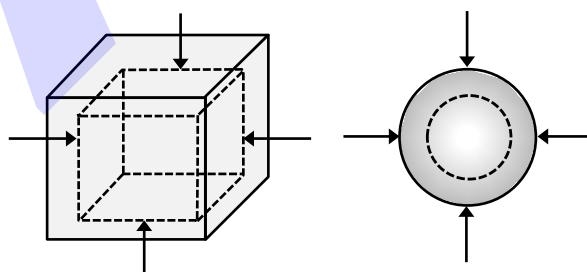
It is a unitless and dimensionless quantity.

There are three types of strain : Type of strain depends upon the directions of applied force.

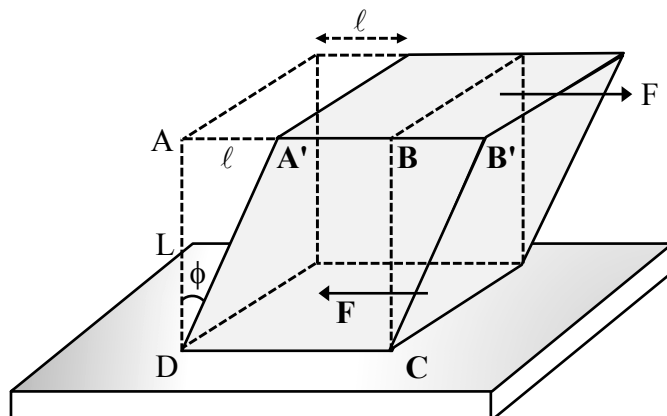
$$\text{Longitudinal strain} = \frac{\text{change in length of the body}}{\text{initial length of the body}} = \frac{\Delta L}{L}$$



$$\text{Volume strain} = \frac{\text{change in volume of the body}}{\text{original volume of the body}} = \frac{\Delta V}{V}$$

**Shear strain**

When a deforming force is applied to a body parallel to its surface then its shape (not size) changes. The strain produced in this way is known as shear strain. The strain produced due to change of shape of the body is known as shear strain.

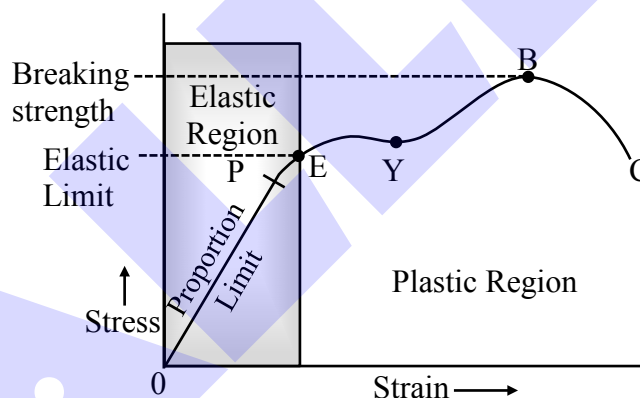


$$\tan \phi = \frac{l}{L} \text{ or } \phi = \frac{l}{L} = \frac{\text{displacement of upper face}}{\text{distance between two faces}}$$

When a material is under tensile stress restoring force are caused by intermolecular attraction while under compressive stress, the restoring force are due to intermolecular repulsion.

If the deforming force is inclined to the surface at an angle θ such that $\theta \neq 0$ and $\theta \neq 90^\circ$ then both tangential and normal stress are developed.

STRESS – STRAIN GRAPH



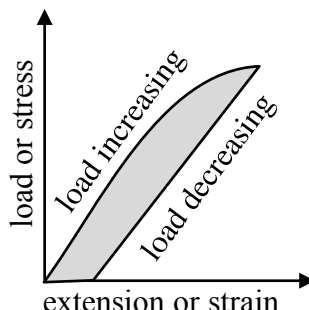
Proportion Limit : The limit in which Hooke's law is valid and stress is directly proportional to strain is called proportion limit. $\text{Stress} \propto \text{Strain}$

Elastic limit : That maximum stress which on removing the deforming force makes the body to recover completely its original state.

Yield Point : The point beyond elastic limit, at which the length of wire starts increasing without increasing stress, is defined as the yield point.

Breaking Point : The position when the strain becomes so large that the wire breaks down at last, is called breaking point. At this position the stress acting in that wire is called breaking stress and strain is called breaking strain.

Elastic Hysteresis : The strain persists even when the stress is removed. This lagging behind of strain is called elastic hysteresis. This is the reason why the values of strain for same stress are different while increasing the load and while decreasing the load.



Breaking Stress : The stress required to cause actual fracture of a material is called the breaking stress

$$\text{Breaking stress} = F/A$$

Breaking stress also measures the tensile strength.

Metals with small plastic deformation are called brittle.

Metals with large plastic deformation are called ductile.

Elasticity restoring forces are strictly conservative only when the elastic hysteresis is zero. i.e., the loading and unloading stress – strain curves are identical.

HOOKE'S LAW

If the deformation is small, the stress in a body is proportional to the corresponding strain, this

fact is known as Hooke's Law. Within elastic limit : $\text{stress} \propto \text{strain} \Rightarrow \frac{\text{stress}}{\text{strain}} = \text{constant}$

This constant is known as modulus of elasticity or coefficient of elasticity.

The modulus of elasticity depends only on the type of material used. It does not depend upon the value of stress and strain.

YOUNG'S MODULUS OF ELASTICITY 'Y'

Within elastic limit the ratio of longitudinal stress and longitudinal strain is called Young's modulus of elasticity.

$$Y = \frac{\text{longitudinal stress}}{\text{longitudinal strain}} = \frac{F/A}{\ell/L} = \frac{FL}{\ell A}$$

BULK MODULUS OF ELASTICITY 'K' or 'B'

Within elastic limit the ratio of the volume stress and the volume strain is called bulk modulus of elasticity.

$$K \text{ or } B = \frac{\text{volume stress}}{\text{volume strain}} = \frac{F/A}{\frac{-\Delta V}{V}} = \frac{\Delta P}{\frac{-\Delta V}{V}}$$

The minus sign indicates a decrease in volume with an increase in stress.

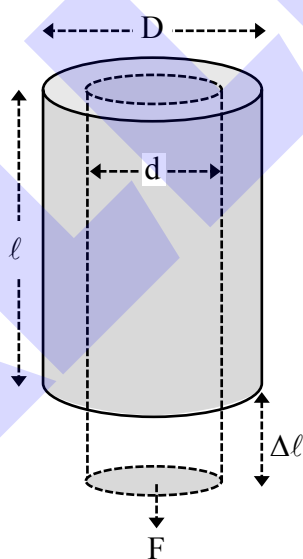
MODULUS OF RIGIDITY 'η'

Within elastic limit the ratio of shearing stress and shearing strain is called modulus of rigidity of a material.

$$\eta = \frac{\text{Shearing stress}}{\text{Shearing strain}} = \frac{\frac{F_{\text{tangential}}}{A}}{\phi} = \frac{F_{\text{tangential}}}{A\phi}$$

Note : Angle of shear 'φ' always taking in radian

Poisson's Ratio (σ)



In elastic limit, the ratio of lateral strain and longitudinal strain is called Poisson's ratio.

$$\sigma = \frac{\text{lateral strain}}{\text{longitudinal strain}} = \frac{\beta}{\alpha}; \beta = \frac{-\Delta D}{D} = \frac{d-D}{D} \text{ \& } \alpha = \frac{\Delta L}{L}$$

EFFECT OF TEMPERATURE ON ELASTICITY

When temperature is increased then due to weakness of inter molecular force the elastic properties in general decreases i.e. elastic constant decreases.

THERMAL EXPANSION

Definition of Heat :

Heat is a form of energy which is transferred between a system and its surrounding as a result of temperature difference only.

Thermal Expansion : Expansion due to increase in temperature.

1. Type of thermal expansion

	Coefficient of expansion	For temperature change Δt	change in
(i) Linear	$\alpha = \lim_{\Delta t \rightarrow 0} \frac{1}{l_0} \frac{\Delta l}{\Delta t}$	length	$\Delta l = l_0 \alpha \Delta t$
(ii) Superficial	$\beta = \lim_{\Delta t \rightarrow 0} \frac{1}{A_0} \frac{\Delta A}{\Delta t}$	Area	$\Delta A = A_0 \beta \Delta t$
(iii) Volume	$\gamma = \lim_{\Delta t \rightarrow 0} \frac{1}{V_0} \frac{\Delta V}{\Delta t}$	volume	$\Delta V = V_0 \gamma \Delta t$
(a) For isotropic solids $\alpha_1 = \alpha_2 = \alpha_3 = \alpha$ (let)			
so $\beta = 2\alpha$ and $\gamma = 3\alpha$			
(b) For anisotropic solids $\beta = \alpha_1 + \alpha_2$ and $\gamma = \alpha_1 + \alpha_2 + \alpha_3$			
Here α_1 , α_2 and α_3 are coefficient of linear expansion in X, Y and Z directions.			

Illustrations

Illustration 1. The difference between lengths of a certain brass rod and of a steel rod is claimed to be constant at all temperatures. Is this possible ?

Solution If L_B and L_S are the lengths of brass and steel rods respectively at a given temperature, then the lengths of the rods when temperature is changed by θ °C.

$$L'_B = L_B(1 + \alpha_B \Delta\theta) \text{ and } L'_S = L_S(1 + \alpha_S \Delta\theta)$$

$$\text{So that } L'_B = L'_S \quad (L_B - L_S) + (L_B \alpha_B - L_S \alpha_S) \Delta\theta$$

So $(L'_B - L'_S)$ will be equal to $(L_B - L_S)$ at all temperatures if, $L_B \alpha_B - L_S \alpha_S = 0$

$$[\text{as } \Delta\theta \neq 0] \text{ or } \frac{L_B}{L_S} = \frac{\alpha_S}{\alpha_B}$$

i.e., the difference in the lengths of the two rods will be independent of temperature if the lengths are in the inverse ratio of their coefficients of linear expansion.

Illustration 2. A steel ruler exactly 20 cm long is graduated to give correct measurements at 20°C.

$$(\alpha_{\text{steel}} = 1.2 \times 10^{-5} \text{ }^{\circ}\text{C}^{-1})$$

- Will it give readings that are too long or too short at lower temperatures?
- What will be the actual length of the ruler be when it is used in the desert at a temperature of 40°C?

Solution

(a) If the temperature decreases, the length of the ruler also decreases through thermal contraction. Below 20°C, each centimeter division is actually somewhat shorter than 1.0 cm, so the steel ruler gives readings that are too long.

(b) At 40°C, the increases in length of the ruler is

$$\Delta \ell = \ell \alpha \Delta T = (20) (1.2 \times 10^{-5}) (40^{\circ} - 20^{\circ}) = 0.48 \times 10^{-2} \text{ cm}$$

The actual length of the ruler is, $\ell' = \ell + \Delta \ell = 20.0048 \text{ cm}$

Illustration 3. A second's pendulum clock has a steel wire. The clock is calibrated at 20°C. How much time does the clock lose or gain in one week when the temperature is increased to 30°C? ($\alpha_{\text{steel}} = 1.2 \times 10^{-5} \text{ }^{\circ}\text{C}^{-1}$)

Solution

The time period of second's pendulum is 2 second. As the temperature increases length time period increases. Clock becomes slow and it loses the time. The change in time period is

$$\Delta T = \frac{1}{2} T \alpha \Delta \theta = \left(\frac{1}{2} \right) (2) (1.2 \times 10^{-5}) (30^{\circ} - 20^{\circ}) = 1.2 \times 10^{-4} \text{ s}$$

$$\therefore \text{New Time period is } T' = T + \Delta T = (2 + 1.2 \times 10^{-4}) = 2.00012 \text{ s}$$

$$\therefore \text{Time lost in one week } \Delta t = \left(\frac{\Delta T}{T'} \right) t = \frac{(1.2 \times 10^{-4})}{(2.00012)} (7 \times 24 \times 3600) = 36.28 \text{ s}$$

Illustration 4. A rod has variable co-efficient of linear expansion $\alpha = \frac{x}{5000}$. If length of the rod is

1m, determine increase in length of the rod in (cm) on increasing temperature of the rod by 100°C.



Ans. 1

Solution Increase in length of $dx = \alpha \ell_0 \delta T = \left(\frac{x}{5000} \right) (dx)(100) = \frac{x}{50} dx$

$$\Rightarrow \text{Total thermal expansion} = \int_0^1 \frac{x}{50} dx = \left(\frac{x^2}{100} \right)_0^1 = \frac{1}{100} \text{ m} = 1 \text{ cm}$$

2. Variation in density : With increase of temperature volume increases so density decreases and vice-versa.

$$d = \frac{d_0}{(1 + \gamma \Delta t)}$$

For solids values of γ are generally small so we can write $d = d_0 (1 - \gamma \Delta t)$ (using binomial expansion)

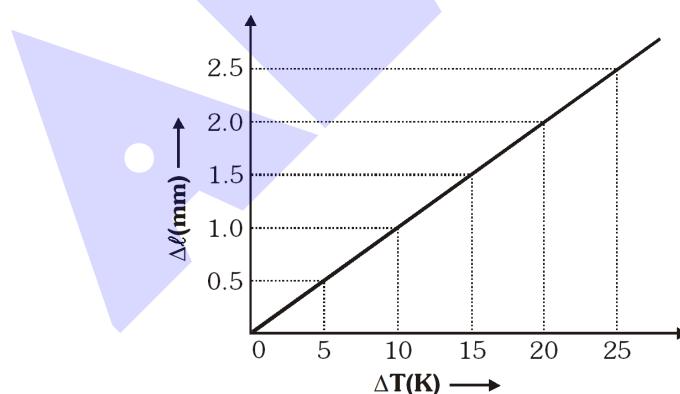
Note :

(i) γ for liquids are in order of 10^{-3}

(ii) For water density increases from 0 to 4°C so γ is -ve (0 to 4°C) and for 4°C to higher temperature γ is +ve. At 4°C density is maximum.

Illustrations

Illustration 5. Figures shows the expansion of a 2m long metal rod with temperature. The volume expansion coefficient of the metal is :



(A) $3 \times 10^{-4} \text{ K}^{-1}$

(B) $1.5 \times 10^{-4} \text{ K}^{-1}$

(C) $3 \times 10^{-5} \text{ K}^{-1}$

(D) $1.5 \times 10^{-5} \text{ K}^{-1}$

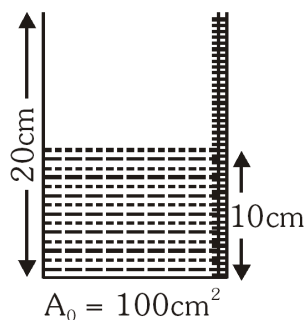
Ans. (B)

Solution $\therefore \alpha = \frac{\Delta \ell}{\ell \Delta T} = \frac{0.5 \times 10^{-3}}{2(5)} = 5 \times 10^{-5} \text{ K}^{-1} \Rightarrow \gamma = 3\alpha = 1.5 \times 10^{-4} \text{ K}^{-1}$

Illustration 6 to 8.

At 20°C a liquid is filled upto 10 cm height in a container of glass of length 20 cm and cross-sectional area 100 cm². Scale is marked on the surface of container. This scale gives correct reading at 20°C.

Given $\gamma_L = 5 \times 10^{-5} \text{ K}^{-1}$, $\alpha_g = 1 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$



6. The volume of liquid at 40°C is :-

- (A) 1002 cc (B) 1001 cc (C) 1003 cc (D) 1000.5 cc

Ans. (B)

Solution

$$V = V_0 (1 + \gamma_L \Delta T) = (10) (100) [1 + 5 \times 10^{-5} \times 20] = 1000 (1 + 0.001) = 1001 \text{ cm}^3 = 1001 \text{ cc}$$

7. The actual height of liquid at 40°C is-

- (A) 10.01 cm (B) 10.006 cm (C) 10.6 cm (D) 10.1 cm

Ans. (B)

Solution

Cross sectional area of vessel at 40°C

$$A = A_0 (1 + 2 \alpha_g \Delta T) = 100 (1 + 2 \times 10^{-5} \times 20) = 100.04 \text{ cm}^2$$

$$\text{Actual height of liquid} = \frac{\text{Actual volume of liquid}}{\text{cross-sectional area of vessel}} = \frac{1001}{100.04} = (1001) (100 + 0.04)^{-1}$$

$$= \left(\frac{1001}{100} \right) \left(1 + \frac{0.04}{100} \right)^{-1} = \frac{(1001)}{100} \left(1 - \frac{0.04}{100} \right) = \frac{1}{100} (1001 - 0.4) = 10.006 \text{ cm}$$

8. The reading of scale at 40°C is-

- (A) 10.01 cm (B) 10.004 cm (C) 10.006 cm (D) 10.04 cm

Ans. (B)

Solution

$$\therefore TV = SR (1 + \alpha_g \Delta T) = (TV) (1 - \alpha_g \Delta T)$$

$$\therefore SR = (TV) (1 + \alpha_g \Delta T)^{-1} = (10.006) (1 - 10^{-5} \times 20) = 10.006 - 0.002 = 10.004 \text{ c}$$

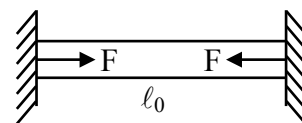
3. **Thermal Stress :** A rod of length ℓ_0 is clamped between two fixed walls with distance ℓ_0 . If temperature is changed by amount Δt then

$$\text{stress} = \frac{F}{A} \quad (\text{area assumed to be constant})$$

$$\text{strain} = \frac{\Delta \ell}{\ell_0}$$

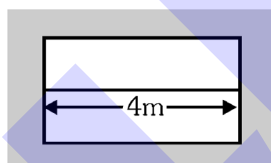
$$\text{so, } Y = \frac{F/A}{\Delta \ell / \ell_0} = \frac{F \ell_0}{A \Delta \ell} = \frac{F}{A \alpha \Delta t}$$

$$\text{or } F = Y A \alpha \Delta t$$



Illustrations

Illustration 9. A fine steel wire of length 4m is fixed rigidly in a heavy brass frame as shown in figure. It is just taut at 20°C . The tensile stress developed in steel wire if whole system is heated to 120°C is :- (Given $\alpha_{\text{brass}} = 1.8 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$, $\alpha_{\text{steel}} = 1.2 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$, $Y_{\text{steel}} = 2 \times 10^{11} \text{ Nm}^{-2}$, $Y_{\text{brass}} = 1.7 \times 10^7 \text{ Nm}^{-2}$)



(A) $1.02 \times 10^4 \text{ Nm}^{-2}$

(B) $1.2 \times 10^8 \text{ Nm}^{-2}$

(C) $1.2 \times 10^6 \text{ Nm}^{-2}$

(D) $6 \times 10^8 \text{ Nm}^{-2}$

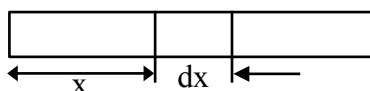
Ans. (B)

Solution Stress = Y (strain) = $Y_s(\alpha_b - \alpha_s) \Delta T = (2 \times 10^{11}) (0.6 \times 10^{-5}) (100) = 1.2 \times 10^8 \text{ Nm}^{-2}$

4. If α is not constant

(i) (α varies with distance) Let $\alpha = ax + b$

$$\text{Total expansion} = \int \text{expansion of length } dx = \int_0^l (ax + b) dx \Delta t$$



(ii) (α varies with temperature)

$$\text{Let } \alpha = f(T)$$

$$\Delta \ell = \int_{T_1}^{T_2} \alpha \ell_0 dT$$

Caution : If α is in $^\circ\text{C}$ then put T_1 and T_2 in $^\circ\text{C}$. Similarly if α is in K then put T_1 and T_2 in K.

CALORIMETRY

Quantity of heat transferred and specific heat

The amount of heat needed to increase the temperature of 1 gm of water from 14.5° C to 15.5° C at STP is 1 calorie

$$dQ = mc dT$$

$$Q = m \int_{T_1}^{T_2} C dT \quad (\text{be careful about unit of temperature, use units according to the given units of } C)$$

Illustrations

Illustration 10. The specific heat of a metal at low temperatures varies according to $S = (4/5)T^3$ where T is the absolute temperature. Find the heat energy needed to raise unit mass of the metal from $T = 1$ K to $T = 2$ K.

Ans. 3

Solution
$$Q = \int m S dT = \frac{m T^4}{5} \Rightarrow \frac{Q}{m} = \frac{15}{5} = 3$$

Heat transfer in phase change

$$Q = mL$$

L = latent heat of substance in cal/ gm/ °C or in Kcal/ kg/ °C

$$L_{\text{ice}} = 80 \text{ cal/ gm for ice}$$

$$L_{\text{steam}} = 540 \text{ cal/ gm}$$

Illustrations

Illustration 11. 5g ice at 0°C is mixed with 5g of steam at 100°C. What is the final temperature?

Solution Heat required by ice to raise its temperature to 100°C,

$$Q_1 = m_1 L_1 + m_1 c_1 \Delta \theta_1 = 5 \times 80 + 5 \times 1 \times 100 = 400 + 500 = 900 = 1800 \text{ cal}$$

$$\text{Heat given by steam when condensed } Q_2 = m_2 L_2 = 5 \times 536 = 2680 \text{ cal}$$

As $Q_2 > Q_1$. This means that whole steam is not even condensed.

Hence temperature of mixture will remain at 100°C.

Illustration 12. Steam at 100°C is passed into 1.1 kg of water contained in a calorimeter of water equivalent 0.02 kg at 15°C till the temperature of the calorimeter and its contents rises to 80°C . What is the mass of steam condensed? Latent heat of steam = 536 cal/g .

Solution Heat required by (calorimeter + water)

$$Q = (m_1 c_1 + m_2 c_2) \Delta\theta = (0.02 + 1.1 \times 1) (80 - 15) = 72.8 \text{ kcal}$$

If m is mass of steam condensed, then heat given by steam

$$Q = mL + mc \Delta\theta = m \times 536 + m \times 1 \times (100 - 80) = 556 m \therefore 556 m = 72.8$$

$$\therefore \text{Mass of steam condensed } m = \frac{72.8}{556} = 0.130 \text{ kg}$$

Illustration 13. The temperature of a body rises by 44°C when a certain amount of heat is given to it. The same heat when supplied to 22 g of ice at -8°C , raises its temperature by 16°C . The water equivalent of the body is

[Given : $s_{\text{water}} = 1 \text{ cal/g}^{\circ}\text{C}$ & $L_f = 80 \text{ cal/g}$, $s_{\text{ice}} = 0.5 \text{ cal/g}^{\circ}\text{C}$]

(A) 25g

(B) 50 g

(C) 80 g

(D) 100 g

Ans. (B)

Solution Supplied heat = $(22) (0.5) (8) + (22) (80) + (22) (1) (16) = 88 + 1760 + 352 = 2200 \text{ cal}$

$$\text{Heat capacity of the body} = \frac{2200 \text{ cal}}{44^{\circ}\text{C}} = 50 \text{ cal/}^{\circ}\text{C}$$

$$\text{Water equivalent of the body} = \frac{\text{Heat capacity of the body}}{\text{specific heat capacity of water}} = \frac{50 \text{ cal/}^{\circ}\text{C}}{1 \text{ cal/g}^{\circ}\text{C}} = 50\text{g}$$

HEAT-TRANSFER

(A) **Conduction** : Due to vibration and collision of medium particles.

(i) **Steady State** : In this state heat absorption stops and temperature gradient throughout the rod

becomes constant i.e. $\frac{dT}{dx} = \text{constant}$.

(ii) **Before steady state** : Temperature of rod at any point changes

Note : If specific heat of any substance is zero, it can be considered always in steady state.

1. Ohm's law for Thermal Conduction in Steady State :

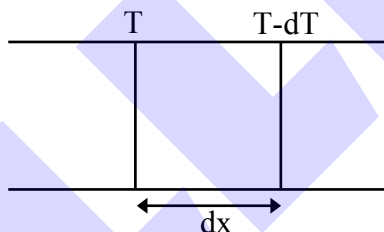
Let the two ends of rod of length ℓ is maintained at temp T_1 and T_2 ($T_1 > T_2$)

$$\text{Thermal current } \frac{dQ}{dT} = \frac{T_1 - T_2}{R_{Th}}$$

Where thermal resistance $R_{Th} = \frac{1}{K} \frac{1}{A}$



2. Differential form of Ohm's Law



$$\frac{dQ}{dT} = KA \frac{dT}{dx}; \frac{dT}{dx} = \text{temperature gradient}$$

Illustrations

Illustration 14. One face of an aluminium cube of edge 2 metre is maintained at 100°C and the other end is maintained at 0°C . All other surfaces are covered by adiabatic walls. Find the amount of heat flowing through the cube in 5 seconds. (thermal conductivity of aluminium is $209 \text{ W/m-}^\circ\text{C}$)

Solution

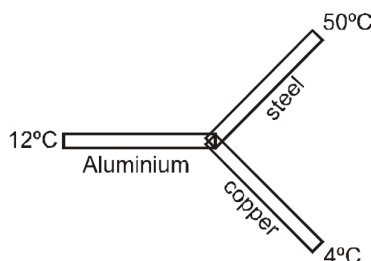
Heat will flow from the end at 100°C to the end at 0°C .

Area of cross-section perpendicular to direction of heat flow, $A = 4\text{m}^2$ then

$$\frac{Q}{t} = KA \frac{(T_H - T_C)}{L}$$

$$Q = \frac{(209 \text{ W/m}^\circ\text{C})(4\text{m}^2)(100^\circ\text{C} - 0^\circ\text{C})(5\text{sec})}{2\text{m}} = 209 \text{ kJ}$$

Illustration 15. Three identical rods of length 1m each, having cross-section area of 1cm^2 each and made of Aluminium, copper and steel respectively are maintained at temperatures of 12°C , 4°C and 50°C respectively at their separate ends. Find the temperature of their common junction. [$K_{\text{Cu}} = 400 \text{ W/m-K}$, $K_{\text{Al}} = 200 \text{ W/m-K}$, $K_{\text{steel}} = 50 \text{ W/m-K}$]



Solution

$$R_{\text{Al}} = \frac{L}{KA} = \frac{1}{10^{-4} \times 200} = \frac{10^4}{200}$$

$$\text{Similarly } R_{\text{steel}} = \frac{10^4}{50} \text{ and } R_{\text{copper}} = \frac{10^4}{400}$$

Let temperature of common junction = T
then from Kirchoff's current laws,

$$i_{\text{Al}} + i_{\text{steel}} + i_{\text{Cu}} = 0$$

$$\Rightarrow \frac{T-12}{R_{\text{Al}}} + \frac{T-51}{R_{\text{steel}}} + \frac{T-4}{R_{\text{Cu}}} = 0$$

$$\Rightarrow (T-12) 200 + (T-51) 50 + (T-4) 400$$

$$\Rightarrow 4(T-12) + (T-51) + 8(T-4) = 0$$

$$\Rightarrow 13T = 48 + 50 + 32 = 130 \Rightarrow T = 10^\circ\text{C}$$

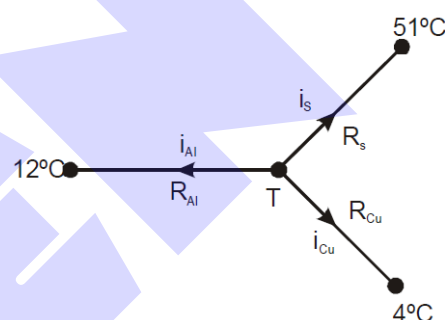
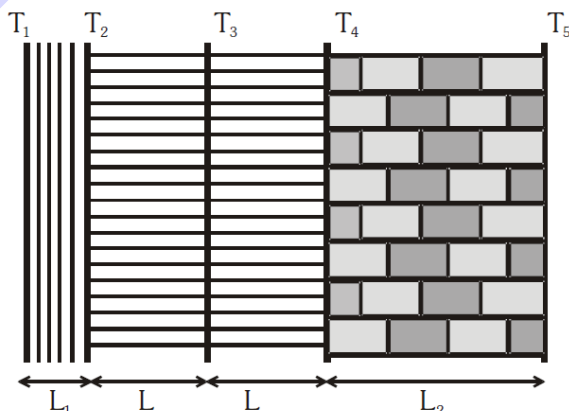


Illustration 16. The figure shows the cross-section of the outer wall of a house built in a hill-resort to keep the house insulated from the freezing temperature of outside. The wall consists of teak wood of thickness L_1 and brick of thickness ($L_2 = 5L_1$), sandwiching two layers of an unknown material with identical thermal conductivities and thickness. The thermal conductivity of teak wood is K_1 and that of brick is ($K_2 = 5K$). Heat conduction through the wall has reached a steady state with the temperature of three surfaces being known.

($T_1 = 25^\circ\text{C}$, $T_2 = 20^\circ\text{C}$ and $T_5 = -20^\circ\text{C}$). Find the interface temperature T_4 and T_3 .

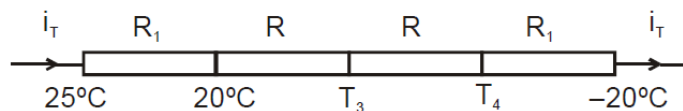


Solution

Let interface area be A then thermal resistance of wood, $R_1 = \frac{L_1}{K_1 A}$ and that of brick

$$\text{wall } R_2 = \frac{L_2}{K_2 A} = \frac{5L_1}{5K_1 A} = R_1$$

Let thermal resistance of the each sand witch layer = R . Then the above wall can be visualised as a circuit



Thermal current through each wall is same.

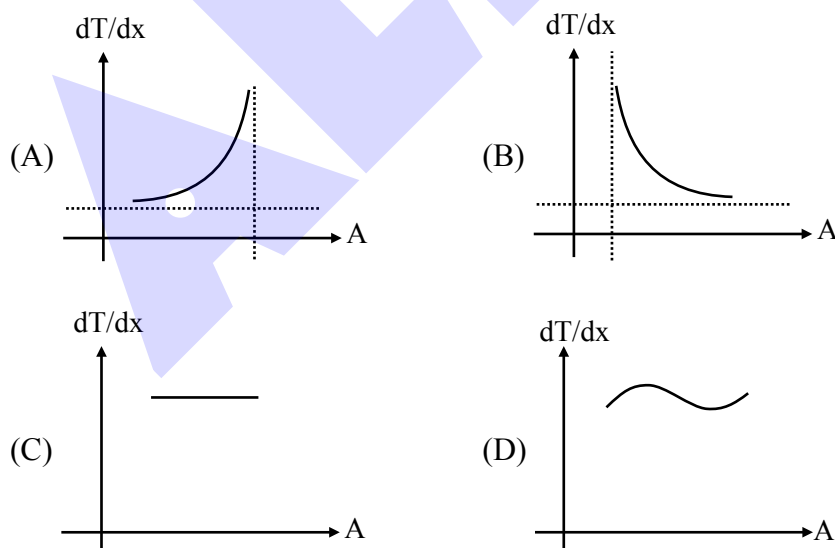
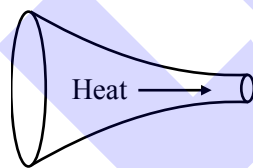
$$\text{Hence } \frac{25 - 20}{R_1} = \frac{20 - T_3}{R} = \frac{T_3 - T_4}{R} = \frac{T_4 + 20}{R_1}$$

$$\Rightarrow 25 - 20 = T_4 + 20 \Rightarrow T_4 = -15^\circ\text{C}$$

$$\text{also, } 20 - T_3 = T_3 - T_4$$

$$\Rightarrow T_3 = \frac{20 + T_4}{2} = 2.5^\circ\text{C}$$

Illustration 17. An irregular rod of same uniform material as shown in figure is conducting heat at a steady rate. The temperature gradient at various sections versus area of cross section graph will be



Ans. (B)

Solution $H = KA \frac{dT}{dx}$ is same in steady state condition, $\therefore A \frac{dT}{dx} = \text{constant}$

$\therefore$ rectangular hyperbolic graph

Illustration 18. Three rods of material X and three rods of material Y are connected as shown in figure. All the rods are identical in length and cross-sectional area. If the end A is maintained at 60°C and the junction E at 10°C , calculate the temp. of the junctions B, C, D. The thermal conductivity of X is 0.92 CGS units and that of Y is 0.46 CGS units.

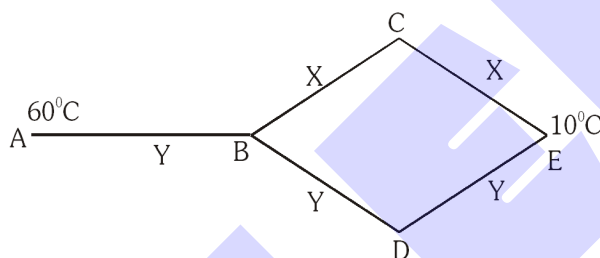
Solution $R_X \propto \frac{1}{K_X}, R_Y \propto \frac{1}{K_Y} \Rightarrow \frac{R_X}{R_Y} = \frac{K_Y}{K_X} = \frac{0.46}{0.92} = \frac{1}{2}$

Let $R_X = R \therefore R_Y = 2R$

The total resistance $\Sigma R = R_Y + \text{effective resistance in the bridge}$

$$\Sigma R = 2R + \frac{2R \times 4R}{2R + 4R} = 2R + \frac{4}{3}R = \frac{10}{3}R \quad \& \therefore \Delta\theta = \ell \times R$$

Further $I_{BCE}(2R) = I_{BDE}(4R)$ and $I_{BCE} + I_{BDE} = I \Rightarrow I_{BCE} = \frac{2}{3}I$ and $I_{BDE} = \frac{1}{3}I$



For A and B $\theta_A - \theta_B = 60^\circ - \theta_B \Rightarrow 60 - \theta_B = 2R \times I \quad \dots(i)$

For B and C $\theta_B - \theta_C = 60^\circ - \theta_B \dots(ii) \quad 60 - \theta_B = \frac{2}{3} \times R \times I$

For A and E $\theta_A - \theta_E = 60 - 10 = 50 \Rightarrow \frac{10}{3} (R \times I) = 50 \dots(iii) \therefore R \times I = 15$

$$\therefore \theta_A - \theta_B - 2 \times 15 = 30, \theta_B = 60 - 30 = 30^\circ\text{C}, \theta_B - \theta_C = \left(\frac{2}{3}\right) \times 15 = 10$$

$$\therefore \theta_C = 30 - 10 = 20^\circ\text{C} \text{ Obviously, } \theta_C = \theta_D = 20^\circ\text{C}$$

Illustration 19. Steam at 373 K is passed through a tube of radius 10 cm and length 2 m. The thickness of the tube is 5 mm and thermal conductivity of the material is $390 \text{ W m}^{-1} \text{ K}^{-1}$, calculate the heat lost per second. The outside temp. is 0°C .

Solution Using the relation $Q = \frac{KA(T_1 - T_2)t}{L}$

Here, heat is lost through the cylindrical surface of the tube.

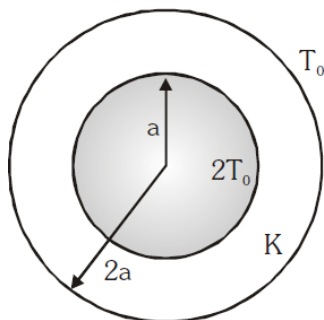
$$A = 2\pi r (\text{radius of the tube}) (\text{length of the tube}) = 2\pi \times 0.1 \times 2 = 0.4 \pi \text{ m}^2$$

$$K = 390 \text{ W m}^{-1} \text{ K}^{-1}$$

$$T_1 = 373 \text{ K}, T_2 = 0^\circ\text{C} = 273 \text{ K}, L = 5 \text{ mm} = 0.005 \text{ m and } t = 1 \text{ s}$$

$$\therefore Q = \frac{390 \times 0.4\pi \times (373 - 273) \times 1}{0.005} = \frac{390 \times 0.4\pi \times 100}{0.005} = 98 \times 10^5 \text{ J.}$$

Illustration 20. A hollow spherical ball of inner radius a and outer radius $2a$ is made of a uniform material of constant thermal conductivity K . The temperature within the ball is maintained at $2T_0$ and outside the ball it is T_0 . Find, (a) the rate at which heat flows out of the ball in the steady state, (b) the temperature at $r = 3a/2$, where r is radial distance from the centre of shell. Assume steady state condition.



Solution In the steady state, the net outward thermal current is constant, and does not depend on the radial position.

$$\text{Thermal current, } C_1 = \left(\frac{dQ}{dt} \right) = -K \cdot (4\pi r^2) \frac{dT}{dr} \Rightarrow \frac{dT}{dr} = -\frac{C_1}{4\pi K} \frac{1}{r^2} \quad T = \frac{C_1}{4\pi K r} + C_2$$

$$\text{At } r = a, T = 2T_0 \text{ and } r = 2a, T = T_0 \Rightarrow T = \frac{2a}{r} T_0$$

$$(a) \frac{dQ}{dt} = 8\pi a K T_0 \quad (b) T(r = 3a/2) = 4T_0/3$$

(A) Convection : Heat transfer due to movement of medium particles.

(B) Radiation: Every body radiates electromagnetic radiation of all possible wavelength at all temp $> 0K$.

(i) Stefan's Law : Rate of heat emitted by a body at temp T K from per unit area $E = \sigma T^4$
J/sec/m² Radiation power $\frac{dQ}{dT} = P = \sigma A T^4$ watt

If a body is placed in a surrounding of temperature T_s : $\frac{dQ}{dT} = \sigma A (T^4 - T_s^4)$ valid only for black body

Emissivity or emissive power (e) = $\frac{\text{heat from general body}}{\text{heat from black body}}$

If temp of body falls by dT in time dt : $\frac{dT}{dt} = \frac{eA\sigma}{mS} (T^4 - T_s^4)$ (dT/dt = rate of cooling)

(ii) Newton's law of cooling

If temperature difference of body with surrounding is small i.e. $\Delta T \ll T_s$ then $\frac{dT}{dt} \propto (T - T_s)$

(iii) General form of Newton's law of cooling

If a body cools from T_1 to T_2 in time δt

$$\frac{T_1 - T_2}{\delta t} = \frac{K}{mS} \left(\frac{T_1 + T_2}{2} - T_s \right) \quad (\text{used generally in objective questions})$$

$$\frac{dT}{dt} = \frac{K}{mS} (T - T_s) \quad (\text{for better results use this generally in subjective})$$

Illustrations

Illustration 21. If a liquid takes 30 seconds in cooling of 80°C to 70°C and 70 seconds in cooling 60°C to 50°C , then find the room temperature.

Solution

$$\frac{\theta_1 - \theta_2}{t} = K \left(\frac{\theta_1 + \theta_2}{2} - \theta_0 \right)$$

$$\text{In first case, } \frac{80 - 70}{30} = K \left(\frac{80 + 70}{2} - \theta_0 \right) \quad \frac{1}{3} = K (75 - \theta_0) \quad \dots (i)$$

$$\text{In second case, } \frac{60 - 50}{70} = K \left(\frac{60 + 50}{2} - \theta_0 \right) \quad \frac{1}{7} = K (55 - \theta_0) \quad \dots (ii)$$

$$\text{Equation (i) divide by equation (ii)} \quad \frac{7}{3} = \frac{(75 - \theta_0)}{(55 - \theta_0)}$$

$$\Rightarrow 385 - 7\theta_0 = 225 - 3\theta_0 \Rightarrow \theta_0 = \frac{160}{4} = 40^\circ\text{C}$$

Illustration 22 to 24.

A body cools in a surrounding of constant temperature 30°C . Its heat capacity is $2\text{J}/^\circ\text{C}$. Initial temperature of the body is 40°C . Assume Newton's law of cooling is valid. The body cools to 36°C in 10 minutes.

22. In further 10 minutes it will cool from 36°C to :

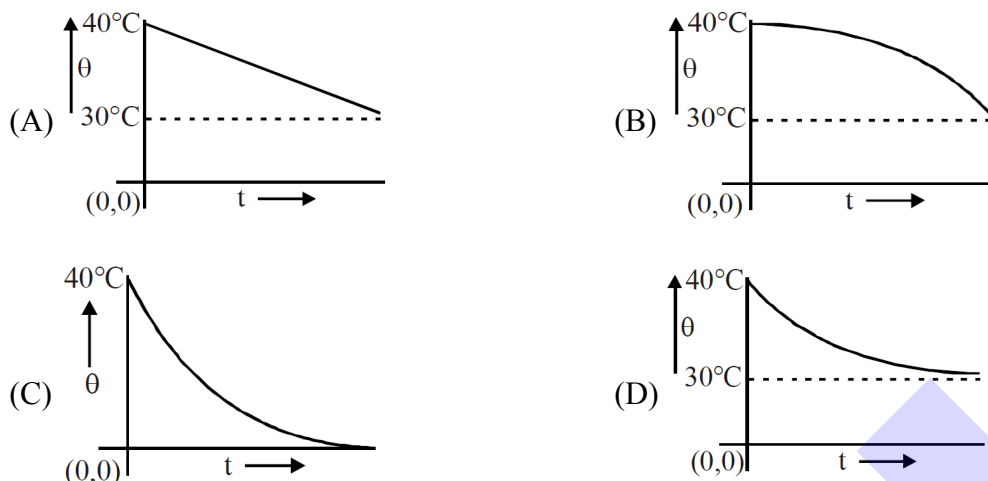
- (A) 34.8°C (B) 32.1°C (C) 32.8°C (D) 33.6°C

Ans. (D)

Solution

$$\frac{40 - 36}{10} = k(38 - 30) \text{ and } \frac{36 - x}{10} = k \left(\frac{36 + x}{2} - 30 \right) \Rightarrow x = 33.6$$

23. The temperature of the body in °C denoted by θ the variation of θ versus time t is best denoted as



Ans. (D)

Solution

$$\frac{d\theta}{dt} = -\frac{kA}{ms}(T - T_0) \text{ Magnitude of slope will decrease with time.}$$

24. When the body temperature has reached 36°C, it is heated again so that it reaches to 40°C in 10 minutes. Assume that the rate of loss of heat at 38°C is the average rate of loss for the given time. The total heat required from a heater by the body is :

- (A) 7.2 J (B) 0.728 J (C) 16 J (D) 32 J

Ans. (C)

Solution

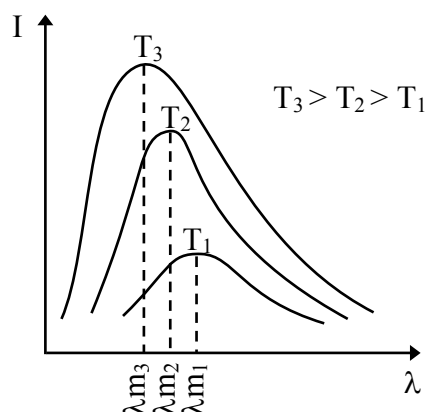
$$\therefore \frac{40-36}{10} = k(38-30) \Rightarrow k = \frac{4}{10 \times 8} = \frac{1}{20}$$

when the block is at 38°C and room temperature is at 30°C the rate of heat loss $ms \times \frac{d\theta}{dt} = ms k (38-30)$

$$\text{Total heat loss in 10 minute} \Rightarrow \Delta Q = ms k (38-30) \times 10 = 2 \times \frac{1}{20} \times 8 \times 10 = 8 \text{ J}$$

Now heat agained by the object in the said 10 minutes. $Q = ms \Delta\theta = 2 \times 4 = 8 \text{ J}$

Total heat required = 8 + 8 = 16 J

(iv) Wein's black body radiation

At every temperature ($> 0\text{K}$) a body radiates energy radiations of all wavelengths.

According to Wein's displacement law if the wavelength corresponding to maximum energy is λ_m .

then $\lambda_m T = b$

where

b = is a constant (Wein's constant)

T = temperature of body

Illustrations

Illustration 25. The temperature of furnace is 2000°C , in its spectrum the maximum intensity is obtained at about $4000\text{\AA}$, If the maximum intensity is at $2000\text{\AA}$ calculate the temperature of the furnace in $^\circ\text{C}$.

Solution By using $\lambda_m T = b$, $4000 (2000 + 273) = 2000(T)$

$$\Rightarrow T = 4546\text{K}$$

$$\text{The temperature of furnace} = 4546 - 273 = 4273^\circ\text{C}$$